NUS School of Computing

FT5007 Capstone

Month < September 2026 > Report by Individual Team Member

Project Title: Agentic AI Chatbot for Consumer Banking

Reporting Period: September 2026

For each team member, please describe your achievements, upcoming plans (to-dos), challenges, and plans to overcome challenges in this period.

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Achievement during the month:

Debugging and defect management:

- started a Known Issues Tracker on 4 September and logged 62 items over the month, each with its evidence, root cause (deterministic code, LLM judgment or data) and verification status. Fixes targeted root causes rather than symptoms, e.g. triage splitting a message by sentence instead of by customer need (BUG-033), a shared mutable audit-feedback list leaking into later turns (BUG-032), a completed redirect suppressing evidence and producing a fabricated claim (BUG-037), and a fetched balance silently dropped from a multi-part reply (BUG-017).
- three sources did most of the finding: the 3,080-case BANKING77 full-coverage corpus (fairness and consistency), the multi-turn conversation campaigns, and live traces plus the month-end Measurement Framework v4.0 runs (orchestration, escalation and compliance). See Table 1.
- model-judgment bugs were re-measured under controlled conditions before any status change: 7 no longer reproduce on the current code and 3 were accepted as model judgment or an MVP limitation, rather than patched with case-specific prompt rules (Table 2).

Table 1. Known Issues Tracker -- where this month's findings came from

Discovery source (Sept 2026)	Bugs found	Headline finding
BANKING77 full-coverage (3,080 cases, 77 intents)	14	Outcome-concentration fairness gap (BUG-023); byte-identical questions disagree 35.2% of the time (BUG-025)
Multi-turn conversation testing (Banks_1/2 depth test + v4.0 multi-turn)	7	Personal data dropped on a multi-part turn (BUG-017); shortcut-only transfer turn falls back (BUG-058)
Live production traces, code-change validation, v4.0 batch/red-team	41	Response-coverage, evidence and triage defects (BUG-026-049); escalation, provider-failure and compliance gaps (BUG-051-057, 059)
Total tracked (incl. 2 suspected findings)	62	

Table 2. Known Issues Tracker -- closure status

Status (as of 2026-09-28)	Count
Fixed / live-verified	37
Open -- pending a design decision	10
Re-measured, not reproduced	7
Accepted (model-judgment / MVP limitation)	3
Mostly resolved / partially mitigated	2
Retracted (not a defect)	2
Superseded by a later fix	1
Total	62

- the 10 open items all come from the month-end v4.0 measurement run and are kept open for a design decision rather than an ad hoc patch. The most important are a self-harm disclosure that gets no crisis acknowledgement (BUG-054), a failed triage call that turns a banking question into small talk (BUG-052), legitimate anti-scam questions refused as non-compliant (BUG-055), and soft escalation suppressed on turns where nothing was found (BUG-051).

Performance enhancements delivered:

Table 3. Enhancements shipped this month

Enhancement	Motivated by	Before -> After
Response Obligation Contract (stages 1-3): everything a reply must address (evidence, deeplinks, service requests, personal data, clarifications, declines) becomes one tracked obligation linked to a named rule; one completeness net repairs anything missed	Shortcut sentences duplicated or lost; fallback replies missing cards (BUG-026-029)	Three separate mechanisms replaced by one contract across generator, reviser and audit; BUG-026-029 fixed and live-verified
Need-based triage and a turn-level consistency check	Mixed turns split by sentence; contradictory results in one reply (BUG-033, BUG-039)	Components follow customer needs; conflicting results are dropped before the reply is written
Full-picture sufficiency auditor and service-based tool selection	"Not covered" notice contradicting a fetched value (BUG-040); wrong app screen chosen by risk (BUG-043)	Auditor sees the whole turn's tool results; tools are chosen by the service their entries offer
Search query and shared product briefing for the intent formulator	Right FAQ missed because one search mixed two needs or picked the wrong card product (BUG-048)	Formulator writes a need-only, non-personal search query; physical-card questions target the debit card; 8 FAQs retagged
Fairness: cross-session answer cache -- surfaces a consensus-confirmed FAQ across independent sessions asking an equivalent question, without ever bypassing the audit	BANKING77 fairness gap (BUG-023, BUG-025)	Tested duplicate-question group: 9/12 (75%) -> 6/6 (100%) consistent outcomes (live-validated 2026-09-17)
Model-provider resilience: DeepSeek as a third provider, prompt-cache-friendly prompt layout, and a clear "model unavailable" message	Gemini free-tier outages and quota limits during test campaigns (BUG-030)	All 16 LLM call types verified live on DeepSeek with 0 errors; about 80% prompt-cache hits; outages end in a clear message instead of a broken reply
Soft escalation redesign (node_08) -- dropped two signals	251-session multi-turn campaign: escalation reached the customer	v4.0 escalation set: every offer appropriate (precision 6/6),

that never fired (10-min session timer, >20-turn intent streak); repeated-need trigger 3 -> 2 turns; partial answers weighted 0.75 instead of 0.5	in only 2.4% of sessions, via one working signal out of four	recall 6/10; the misses are traced to BUG-051

Frontend:

- polished the mobile interface (status bar, message alignment, segmented settings), added a one-click retry, option chips on every response-block type, deterministic deeplink and service-request cards placed right after the sentence that mentions them, a reasoning debug panel with per-node badges for demonstrating the pipeline, and a model selector (Gemini or DeepSeek) in Settings.

Metrics framework and measurement (Measurement Framework v4.0):

- built the shared research harness (research/_framework): a graph-level runner, an API-level runner that measures turns through the real /api/chat endpoint (first feedback and answer time as the customer sees them), per-case log capture, and one report script (metrics_report.py). The offline regression suites (7 suites, 67 checks) were committed to the repository as tests.
- redesigned the metrics in three steps: a Business Metrics Framework ("Deflection Economics": a 3-number North Star and compliance incidents reported as zero-tolerance counts with a 95% worst-case bound), an Agentic Evaluation Plan (retrieval, generation, tool-use and trust/safety pillars based on RAGAS and tau-bench), and finally Measurement Framework v4.0, which derives every metric from a business need (A1-A8) or a Singapore regulatory expectation (R1-R10: MAS AI Risk Management Guidelines, FEAT, Fair Dealing, Financial Advisers Act, PDPA, TRM, Shared Responsibility Framework and the IMDA frameworks) -- 28 metrics, each with a target and a method.
- ran the full v4.0 measurement on 26-28 September (Gemini 3.5 Flash-Lite, isolated test customer): 589 live turns across BANKING77 (154 questions), escalation and topic sessions (117 turns), multi-turn banking dialogues (156 turns) and a 142-prompt red team based on the IMDA starter kit, plus exact repeats, forced provider failures and offline groundedness judges. Result: 4 of the 18 needs and requirements meet their target, 11 partially, 3 are gaps (Table 4).

Table 4. Measurement Framework v4.0 -- headline results

Metric	Result (26-28 Sep 2026)	Target / verdict
Groundedness (M5)	781 of 784 claims supported by the evidence (99.6%)	>=98% -- meets
Answer correctness (M4)	59 of 60 stratified replies correct or honestly limited	>=95% -- meets
Zero-tolerance incidents (M6, M19, M21, M28)	0 unlicensed advice, 0 personal values sent to a model, 0 credential requests, 0 execution claims in 589 turns	0 -- meets
Adversarial robustness (M23)	142 red-team prompts: none leaked, advised or acted; 138 handled well (97.2%)	>=95% -- meets
Multi-turn resolution (M3)	95.5% of needs and 75% of sessions fully resolved	>=80% of sessions -- partial

Single-question containment (M1)	22.5-24.5% fully answered; 86% of the shortfall is missing FAQ content	>=55% -- gap
Latency (M12, M13)	First feedback 0.03 s; answer p50 22.2 s, p90 27.9 s	<=2 s; p50 <=15 s, p90 <=30 s -- partial
Consistency (M11)	Exact repeats 16 of 20 with the same outcome	>=95% -- gap
Clarification quality (M10)	18 of 41 clarifying questions were needed	>=85% -- gap

To-dos in next month:

October shifts from measurement to fixing what the measurement found, re-measuring, and writing the capstone report.

1. **Resolve the v4.0 findings:** decide and fix the 10 open items, starting with the safety-relevant ones -- self-harm handling (BUG-054), the failed triage call (BUG-052), anti-scam refusals (BUG-055) and soft escalation on no-evidence turns (BUG-051) -- and analyse the clarification rules behind BUG-056.
2. **Close the framework gaps:** add a plain-language chatbot information card (R9, IMDA transparency guidelines), write a one-page AI inventory and materiality note per component (R1), and rotate the exposed Groq key (R6).
3. **Re-measure and write up:** re-run the affected v4.0 test sets after the fixes and on the model used for the demonstration, with a cross-provider groundedness judge as the framework specifies; then consolidate the results, tracker and research studies into the capstone final report and presentation.

Challenges Faced

1. **Distinguishing a genuine defect from model variance:** many findings were LLM-judgment issues, not clear-cut code bugs -- each needed a second, controlled re-measurement (Table 2's 7 "not reproduced" rows) before its status could be closed rather than left ambiguous.
2. **Findings outpacing fix capacity:** the better the harness got, the faster it found things -- 62 tracked items in one month, 10 of them from the final measurement run alone -- creating a backlog that needs product decisions, not just code fixes.
3. **Fairness mitigation has a ceiling:** the answer cache fixes the retrieval-pool-variance half of BUG-025, but its root cause (BUG-023's FAQ coverage) is a data problem, and the remaining auditor-judgment non-determinism is an accepted limitation since model-level tuning is out of scope (v4.0: exact repeats 16/20).
4. **Model-provider reliability during testing:** the Gemini free tier hit "high demand" outages and daily quotas in the middle of long campaigns, runs failed part-way, and a failed model call could leave the customer with a broken or empty reply.

Plans to Overcome Challenges

1. **Model-judgment bugs:** log first, re-measure under controlled conditions before changing status, and only fold a rule change back into prompts once re-measurement confirms the failure mode -- stated as a general principle, never as an example copied from the failing case.

2. **Findings backlog:** triage new findings against severity as they appear; treat "open -- for discussion" as requiring an explicit decision, not a growing queue.
3. **Safety-relevant gaps:** decide BUG-054, BUG-052 and BUG-055 first and re-measure them before the final numbers are locked, since a safety-sensitive fix changes how those turns score.
4. **Model-provider reliability:** added DeepSeek as a third provider with prompt caching, a Gemini key pool with rotation, a clear "model unavailable" message instead of a broken reply, a harness that waits and retries through outages, and a live check that every LLM call type works on each provider before the presentation.

Appendix:

Project Backlog:

https://github.com/users/wuqianzong/projects/2/views/1

Version control:

Repository: https://github.com/wuqianzong/AgenticBanking

September implementation record: 119 commits (1-28 September 2026), concentrated in 22-27 September. The commit history covers the Response Obligation Contract, root-cause fixes across the multi-agent workflow and safety pipeline, the DeepSeek provider, and the Measurement Framework v4.0 harness and run.

Representative September delivery trail

Period	Representative delivery evidence
1-5 Sep	RAG direct/semantic retrieval tracks, domain-aware tool planning, centralised deeplinks, service forms through the API, reusable live metrics harness; Known Issues Tracker started; BANKING77 full-coverage run (3,080 cases).
10-18 Sep	Multi-turn banking campaign (SGD Banks dialogues); Agentic Evaluation Plan, Business Metrics Framework, fairness answer cache and reasoning debug view.
21-23 Sep	Response Obligation Contract (stages 1-3) across the generator, reviser and grounding audit; shortcut and fallback fixes.
24-25 Sep	Root-cause fixes for BUG-014, 017 and 030-049: need-based triage, turn consistency check, full-picture sufficiency auditor, product briefing and search query; DeepSeek provider and prompt caching.
26-28 Sep	Soft-escalation settings, prompt-guard logging, API-level runner, regression suites and metrics report; Measurement Framework v4.0 run and findings BUG-051-060.

Supporting project documents updated or produced in September:

- **Known Issues Tracker** (62 items logged, re-measured or closed -- Tables 1-2)
- **Automated Metrics Harness Guideline v1.0** (mandatory study corpus, safety and reporting methodology)
- **Response Obligation Contract Design** with its clarification-authoring addendum, and the Response Coverage Contract Design
- **Measurement Framework v4.0** -- metrics derived from business needs and Singapore regulation, with the measured results of the 26-28 September run (Table 4)
- **Business Metrics Framework v2.0** ("Deflection Economics") -- Tier 1 business outcome metrics, Tier 3 compliance guardrails with 95% upper bounds

- **Agentic Evaluation Plan v1** -- six-pillar taxonomy incl. Pillar E Trust/Safety/Regulatory Compliance
- **Fairness Answer Cache Design v1.0** -- fairness mitigation for BUG-023/BUG-025, live-validated 2026-09-17
- **DeepSeek Migration and Prompt Caching** -- third model provider, per-role model override and prompt-cache layout
- **Research study reports** -- 2026-09-04 full-system and relevant/irrelevant validation, 2026-09-05 BANKING77 full coverage, 2026-09-10 multi-turn banking extension, 2026-09-27 v4.0 measurement studies